Ivy Matobori

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SKILLS

- **Embedded Systems & Firmware:** Embedded C/C++, FreeRTOS, ESP-IDF, ESP32, bare-metal development, peripheral drivers, calibration procedures, flow and temperature measurement
- **Industrial Automation:** PLC programming, ladder logic, Siemens TIA Portal, HMI configuration, machine control, alarm handling, safety interlocks
- **Interfaces & Connectivity:** I^2C, SPI, UART, CAN bus, RS485, Modbus, one-wire, ADC, PWM, GPIO, LoRa, BLE, NFC, WiFi, 4G/LTE
- **Hardware & PCB Design:** Altium Designer, KiCad, Fusion 360, schematic capture, multi-layer PCB layout, MCU & component selection, DFM, PCBA, board bring-up
- **Power & Validation:** Power architecture, regulation, protection, redundancy, low-power design, energy metering, oscilloscope, logic analyzer, system validation, Git, technical documentation

EDUCATION

Jomo Kenyatta University of Agriculture and Technology (JKUAT) — **Nairobi, Kenya**
Bachelor of Science in Control and Instrumentation Engineering — *2019 – 2024*

WORK EXPERIENCE

Quepay Limited, Kenya | ***C++ · FreeRTOS · ESP-IDF · Altium · LTE/BLE/I^2C*** — **Feb 2025 – Present**
Hardware Designer & Technical Operations Lead — *On-site, Nairobi, Kenya*

- Sustained a **99%** device uptime SLA across **KES 10M+** in processed transactions, as measured by fleet monitoring, by building the C++ and FreeRTOS firmware integrating M-Pesa, MTN MoMo, and Airtel Money flows over LTE.
- Consolidated **6+** machine verticals onto one controller board, from water and milk ATMs to vending and laundry, by designing the hardware and **4-layer** PCB in Altium Designer, scaling platform volume to **KES 2M+** in **4 months**.
- Specified, procured, installed, and commissioned controller and sensing hardware across **6** field stations, as measured by site acceptance sign-off, validating each installation on site before hand-off to the operator.
- Cut board-level defect rate to **2%** across **300+** units shipped and per-unit validation time to **12 minutes**, by standardising test procedures and leading a **4-person** team across assembly, inspection, and field deployment.
- Drove long-term measurement accuracy on milk ATM dispensers, holding dispensed volume within **±2%** of setpoint, by developing **calibration procedures** for flow and temperature instrumentation and troubleshooting instrument failures in the field.
- Held dispense faults under **1%** across **250** deployed units, by building fault detection, alarm handling, and safety interlocks into the controller firmware, including over-dispense cutoff and automatic payment reversal on incomplete dispense.
- Reduced field return rate **60%** across **250** installed units, as measured over **12 months** of deployment, by performing board bring-up and signal integrity checks on every revision and root-causing **15** production escalations.

Veno Autobotics, Kenya | ***Level & Energy Metering · PCB Design · LoRa/BLE/NFC*** — **Apr 2024 – Jan 2025**
Embedded Systems Engineer, Embedded Hardware Design — *On-site, Nairobi, Kenya*

- Shipped **7** IoT products across school, utility, retail, and municipal deployments, by owning embedded hardware design on a **5-engineer** studio team spanning level sensing, energy metering, amusement metering, and public display systems.
- Instrumented liquid level and energy consumption for field deployments, by designing the sensing hardware behind Veno's Level Sensor and VeNode energy meter, scaling metering from a single appliance to a full industrial complex.
- Extended device reach beyond cellular coverage across **4** radio technologies, LoRa, BLE, NFC, and WiFi, by designing low-power RF hardware and selecting the radio per deployment against range, power budget, and site constraints.
- Laid out the smallest PCB in company history at **15 mm** by **20 mm**, by taking client hardware from schematic and layout through prototype assembly to enclosure and packaging design with mechanical engineers.
- Held control loop jitter under **2 ms** on **32 KB** of flash, by developing bare-metal C/C++ firmware implementing **8** peripheral drivers over I^2C, SPI, and UART and running end-to-end system verification before design freeze.

Kenya Space Agency, Kenya | ***EPS · Power Distribution · MCU Selection*** — **Jan 2023 – Mar 2023**
Satellite Systems Engineer (EPS) — *On-site, Nairobi, Kenya*

- Achieved **92%** conversion efficiency against a **5 W** orbit-average power budget, as measured by bench characterisation, by designing the satellite power distribution PCB on a **6-layer** stack-up with **4** regulated rails on redundant paths.
- Cut standby current draw **18 mA** against the baseline design, as measured by bench draw, by leading microcontroller selection for the EPS subsystem and trading **6** candidate parts against power, radiation tolerance, and reliability constraints.
- Carried the EPS design through **2** design reviews within a **3-month** placement, as measured by review sign-off, by specifying a redundancy architecture with **6** fault-isolated branches and producing **40** pages of interface and test specification.

Gearbox, Kenya | ***KiCad · Embedded C · Prototyping · Power Management*** — **Jan 2022 – Apr 2022**
Embedded Systems Apprentice — *On-site, Nairobi, Kenya*

- Delivered **3** working embedded demonstrators in **16 weeks**, as measured against **10** functional acceptance criteria per build, by prototyping microcontroller, sensor, and actuator systems and validating each before hand-off.
- Turned **4** boards from concept to tested hardware within a **3-week** cycle, as measured by fabrication turnaround, by completing schematic capture, **2-layer** PCB layout, and DFM review in KiCad.
- Brought **6** sensor interfaces online and cut prototype idle power draw **35%**, as measured by bench current draw, by developing bare-metal firmware in embedded C and optimising supply rails and sleep modes across **4** boards.